Appendix

This appendix presents the electrical models, resource parameters, task construction, benchmark configurations, and additional results for the coordinated operation of distribution networks and microgrids. The experiments comprise 60 core dispatch tasks and 40 tasks with newly introduced operating requirements on 33-bus and 69-bus feeders. The two task sets evaluate computational performance as the resource population increases and task adaptation following rule updates, respectively.

A. Test Systems and Resource Connections

The feeders are represented by balanced AC network models with constant-power (PQ) loads. Line impedances and base loads for the 33-bus system are taken from MATPOWER case33bw [1]; the 69-bus system uses the published line and load data in [2]. Both systems have a nominal voltage of 12.66 kV and a power base of 10 MVA. Bus 1 is the reference bus, with its voltage fixed at $1.00\angle 0°$ p.u. The locations of the points of common coupling (PCCs) are shown in Fig. A1. Bus numbering starts at 1.

TABLE A1
FEEDER PARAMETERS AND PCC CONNECTION LOCATIONS

Parameter	33-bus system	69-bus system
Base active/reactive load	3.715 MW / 2.300 Mvar	3.80189 MW / 2.69410 Mvar
Radial branches	32	68
PCC installation buses	6, 14, 18, 25, 30	11, 18, 27, 35, 46, 49, 59, 61, 65
Base branch current limits	300 A	200 / 300 / 400 A, as specified by branch
Voltage limits / dispatch interval	0.95–1.05 p.u. / 0.25 h	0.95–1.05 p.u. / 0.25 h

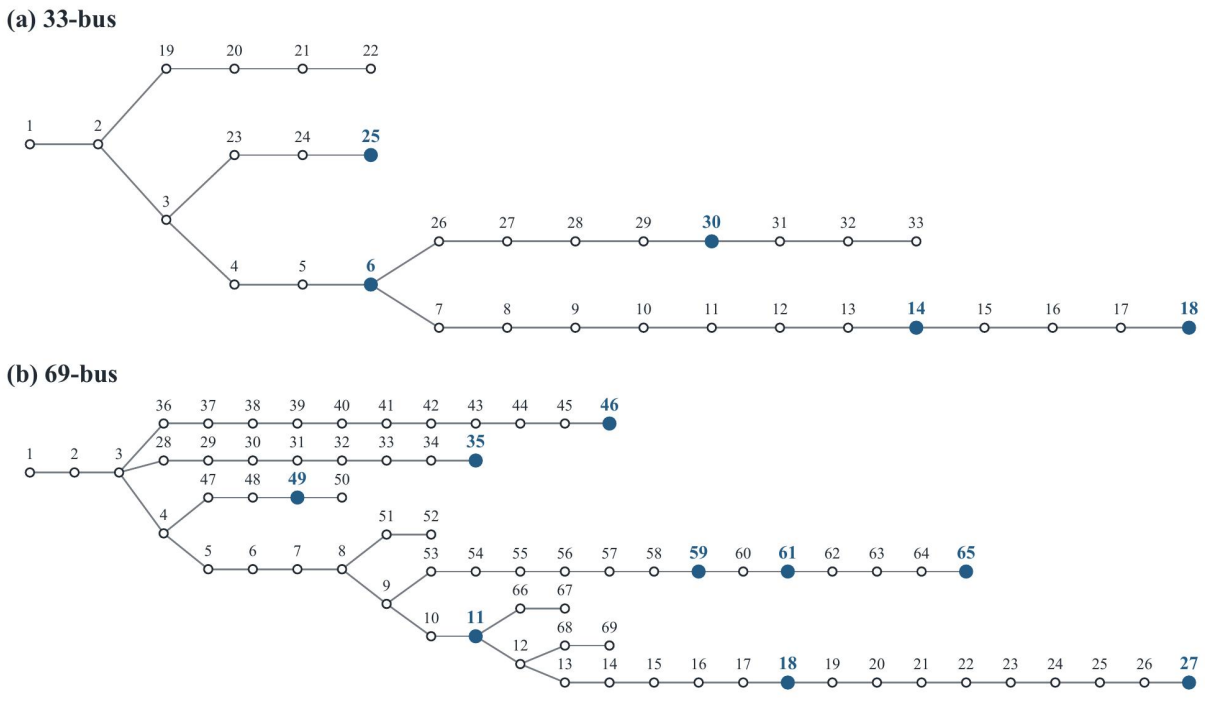


Fig. A1. Feeder topologies and PCC installation buses. Blue nodes indicate installation buses; at the scale factor of 100, each installation bus hosts 100 PCC groups.

The branch current limit for the 33-bus system is set to 300 A. Current limits for the 69-bus system follow the published data table. Base feeder loads and the additional microgrid loads both enter the nodal power balance.

B. Device Constraints and Dispatch Cost

Each PCC group represents a microgrid containing three resource units under a lumped-parameter model. Each unit comprises photovoltaic (PV) generation, energy storage, reactive power support, and flexible load. Its four continuous control variables are storage active power b, reactive power support q, load shedding d, and PV curtailment c. Positive storage power denotes discharge, positive reactive power denotes injection, and positive PCC active power denotes net import. Resources within a group are connected to the same bus, and their powers are summed to obtain the group exchange [4]. Feeder losses are calculated through AC power flow.

TABLE A2
AGGREGATE RESOURCE RATINGS OF THE TWO SYSTEMS

Parameter	33-bus system	69-bus system
Rated PV active power	5.6 MW	8.0 MW
Rated storage power / energy	2.4 MW / 6.0 MWh	2.8 MW / 7.0 MWh
Rated reactive power support	3.0 Mvar	3.4 Mvar
Rated load-shedding capability	0.7 MW	0.8 MW
Additional-load base	1.0 MW	1.2 MW
SOC range / charging and discharging efficiencies	0.20–0.90 / 0.95, 0.95	0.20–0.90 / 0.95, 0.95

Actual resource loads are obtained by multiplying the additional-load base by the scenario load factor. Available PV output is determined by the PV availability factor. The active and reactive power exchanges of PCC group g, and the stored energy of resource k at the end of the dispatch interval, are given by

$$P_g^{\text{in}} = \sum_{k\in G_g}(L_k - P_k^{\text{PV}} - b_k - d_k + c_k),$$
$$Q_g^{\text{in}} = \sum_{k\in G_g}(0.38L_k - q_k), \tag{A1}$$

$$E_k^{+} = E_k s_k - \Delta t\left(\frac{[b_k]_+}{\eta_k^{\text{dis}}} + \eta_k^{\text{ch}}[b_k]_-\right),$$
$$[b]_+ = \max(b,0), [b]_- = \min(b,0). \tag{A2}$$

Here, E is rated storage energy and s is state of charge (SOC). The ratings B, Q, and D denote storage active power, reactive power support, and load-shedding capability, respectively; S is the inverter apparent power rating, and L is the resource load. Storage power bounds depend jointly on the power rating and the available energy headroom. The load-shedding limit is the smaller of local demand and rated shedding capability. PV active and reactive power are coupled by the inverter capability circle. Storage is represented by a separate active power port.

$$-\min\left(B_k, \frac{(s_k^{\max}-s_k)E_k}{\Delta t\eta_k^{\text{ch}}}\right) \le b_k \le \min\left(B_k, \frac{(s_k-s_k^{\min})E_k\eta_k^{\text{dis}}}{\Delta t}\right),$$
$$0 \le d_k \le \min(D_k, L_k), 0 \le c_k \le P_k^{\text{PV}},$$
$$(P_k^{\text{PV}} - c_k)^2 + q_k^2 \le S_k^2, S_k = 1.10P_k^{\text{rated}}. \tag{A3}$$

Each group has a binary mode z that enables reactive power support. Reactive power commands within the group are zero when this mode is disabled. Communication status, control permissions, and device availability determine the corresponding control bounds. All methods use the dispatch cost

$$J = \sum_k[w_{b,k}\left(\frac{b_k^2}{B_k} + 0.02|b_k|\right) + w_{q,k}\frac{q_k^2}{Q_k}$$
$$+ w_{d,k}\left(d_k + 0.1\frac{d_k^2}{D_k}\right) + w_{c,k}\left(c_k + 0.1\frac{c_k^2}{P_k^{\text{rated}}}\right)] + 0.004\sum_g z_g. \tag{A4}$$

The cost function is evaluated using numerical power values expressed in MW and Mvar and serves as a common normalized control-cost measure. The weights incorporate the corresponding cost scaling. The coefficient 0.004 is the reactive-support activation cost per group.

The nodal AC power balance and branch currents use the same per-unit base. The network equations are

$$V_i(YV)_i^* + s_i^{net} = 0, i = 2, \ldots, n,$$
$$0.95 \le |V_i| \le 1.05, |I_e| \le I_e^{\max}, I_e = \frac{V_a - V_b}{z_e}. \quad \text{(A5)}$$

Here, Y is the nodal admittance matrix, s_i^{net} is the per-unit net demand obtained by combining background load and microgrid exchanges, and V is the complex bus voltage. Branch currents and their limits are converted using the current base corresponding to 10 MVA and 12.66 kV.

C. Resource Scaling and Parameter Allocation

The scale factor r takes the values 1, 10, and 100. The 33-bus and 69-bus systems start with five and nine PCC groups, respectively, with three resource units per group. Scaling preserves the feeder topology, background nodal loads, and aggregate resource ratings at each installation area, while increasing the number of PCC groups, individual device states, and control variables. The resulting task dimensions are listed in Table A3.

TABLE A3
CORE TASK DIMENSIONS AND NUMBERS OF CONTINUOUS CONTROLS OPTIMIZED

Buses	r	PCC groups	Resource units	Control entries	Binary modes	Controls in NLP	Controls in proposed method
33	1	5	15	60	5	55–60	55–60
33	10	50	150	600	50	550–600	550–600
33	100	500	1500	6000	500	5500–6000	5500–6000
69	1	9	27	108	9	98–108	98–108
69	10	90	270	1080	90	980–1080	935–1080
69	100	900	2700	10800	900	9800–10800	8100–9450

The last two columns give the observed ranges of the numbers of controls included in continuous optimization. Each resource unit has four continuous control entries. In the normal-operation tasks, zero control attains the zero-cost lower bound.

Let K_a denote an aggregate rated parameter for installation area a. After scaling, this parameter is allocated among PCC groups using positive shares α. The allocated quantities comprise load, PV capacity, storage power and energy, reactive power capacity, shedding capability, and inverter rating. The area-level PCC exchange bounds are divided equally among the r groups.

$$K_{a,m} = \alpha_{a,m} K_a, \alpha_{a,m} > 0,$$
$$\sum_{m=1}^{r} \alpha_{a,m} = 1, m = 1, \ldots, r. \quad \text{(A6)}$$

The base PCC exchange intervals are [−1.3/r, 1.0/r] MW for active power and [−0.8/r, 0.8/r] Mvar for reactive power. In scenarios 004, 005, and 009, the active power import limit is tightened to 0.09/r MW in alternating installation areas.

For all 20 feeder–base-scenario combinations, aggregate resource ratings at each installation area are identical across the three scales. In the 69-bus system, each of the nine installation buses hosts 100 PCC groups, yielding 900 PCC groups, 2,700 resource units, and 900 binary modes for reactive power support. Net exchanges are aggregated at each installation bus and enter the AC power balance in (A5). Bus voltages, branch currents, and losses in the common feeder couple the PCC groups. The three resource units within each group jointly satisfy the PCC exchange bounds and individually satisfy SOC, inverter-rating, and control-permission constraints. This configuration evaluates joint dispatch of multiple participants on a given feeder.

TABLE A4
RESOURCE STATES AND CONTROL-COST PARAMETERS

Parameter	Construction or value	Role in the model
Initial storage SOC	Base samples: 0.42–0.75; restricted units: 0.205	Determines charging and discharging capability
Base cost weights	Storage: 0.8–1.2; reactive power: 0.04; load shedding: 5–7; PV curtailment: 4	Sets the economic priority of control actions
Heterogeneity after subdivision	Bounded perturbations of allocation shares and costs	Creates distinct resource decisions
Preference for preserving stored energy	Storage weight ×15; load-shedding weight ×0.35	Evaluates preference updates at the same physical state
Reactive-support mode	One binary variable per PCC group; activation cost: 0.004	Jointly determines enabled groups and continuous powers

The scaling experiment increases the control dimension and the number of independent PCC decisions while incorporating heterogeneity in resource states and cost weights. The fixed activation cost remains 0.004 per group, so the total potential activation cost increases with the number of groups. All methods use the same objective function for each individual task.

The 60 core tasks combine two feeders, three resource scales, and ten scenario types. The scenario-generation seed is 83059. The ten base states for tasks with new operating requirements are generated using seed 83071; their configurations are specified in Section E.

D. Core Scenarios and Operating Stress Conditions

The core scenarios cover normal operation, voltage deviations, tighter branch current limits, tighter PCC exchange bounds, restricted resource capabilities, and changes in dispatch preferences. Each of the ten scenario types is instantiated once for each feeder and scale, giving 60 tasks. The stress settings and corresponding initial voltage and current conditions are summarized in Table A5.

TABLE A5
TEN CORE SCENARIO TYPES AND INITIAL CONDITIONS WITH ZERO CONTROL

ID	Scenario	Stress setting	Voltage extrema (p.u.)	Range of maximum current ratios
000	Normal operation	Load factor: 0.72; PV availability: 0.48	0.9701–1.0103	0.263–0.321
001	Undervoltage	Higher loading and lower PV output	0.8858–1.0000	0.663–0.828
002	High PV output	Lower loading; PV availability: 0.96–1.00	0.9975–1.0797	0.480–0.536
003	Tighter current limits	Branch current limits multiplied by 0.63	0.8912–1.0000	0.985–1.258
004	Tighter PCC bounds	Tighter active import limits in alternating installation areas	0.8895–1.0000	0.602–0.804
005	Coupled constraints	Current limits and PCC bounds tightened jointly	0.8901–1.0000	1.017–1.292
006	Restricted capabilities	Selected SOCs near the lower bound; additional permission and capability constraints	0.8875–1.0000	0.583–0.805
007	Economic preference	Base cost weights	0.8858–1.0000	0.652–0.835
008	Stored-energy preservation	Same physical state as 007, with updated cost weights	0.8858–1.0000	0.652–0.835
009	Combined stress	Current, PCC, and capability constraints combined	0.8868–1.0000	0.957–1.283

For each scenario type, the voltage column spans the minimum to the maximum bus voltage across its six tasks. The current ratio is the largest branch current divided by its applicable limit in each task; the reported range covers the six tasks. Scenario IDs correspond to the final three digits of the task identifier. Scenarios 007 and 008 form a preference pair with the same physical state.

In the tighter-current-limit scenarios, all branch current limits are multiplied by 0.63. The initial maximum current ratios across the six tasks range from 0.985 to 1.258; the corresponding range for the coupled-constraint scenarios is 1.017–1.292. The tighter PCC exchange intervals introduce additional coupling during subsequent control adjustments; the initial exchanges under zero control lie within these intervals. Voltage, branch current, PCC, and device constraints are enforced jointly throughout dispatch.

Background loads are scaled by the scenario load factor and nodal perturbations. Additional resource loads use the same scenario factor. PV availability is mapped from publicly available hourly profile data. These profiles are derived from the grid-import, grid-export, and cumulative PV energy columns for DE_KN_residential3 in the OPSD Household Data package [3]. Differences between consecutive valid hourly readings yield average powers. Screening for consecutive timestamps, nonnegative increments, and interpolation flags retains 11,174 profile samples.

The public profiles modulate the operating stress conditions. Together, the loads, PV outputs, and resource ratings define the synthetic operating states. Each dispatch decision covers a 0.25 h interval and includes the associated storage energy update. Preference pairs share the same electrical state and are grouped by their common physical parent state during statistical resampling.

E. New Operating Requirements and Task Model Updates

Five base states are selected for each of the 33-bus and 69-bus systems at a scale factor of 100, corresponding to 500 and 900 PCC groups, respectively. Four tasks are constructed from each base state: export limitation, energy retention, fault recovery with load priority, and combined requirements. This yields 40 tasks. Each task is initialized independently from its current state, and the PCC targeted by the new rules is designated group 0.

The four rule types act on the resource set G_0 of the target PCC group, as follows. Each threshold is calculated from the ratings or stored energy of the current base state and is included explicitly in the typed rule.

$$P_0^{\text{in}} \geq -P_{\lim}^{\exp}, P_{\lim}^{\exp} = 0.10 \sum_{k \in G_0} P_k^{\text{PV}}, \tag{A7}$$

$$\sum_{k \in G_0} E_k^{+} \geq E^{\text{req}}, E^{\text{req}} = 1.015 \sum_{k \in G_0} E_k s_k, \tag{A8}$$

$$\sum_{k \in G_0} d_k \leq D^{\text{pri}}, D^{\text{pri}} = 0.05 \sum_{k \in G_0} L_k, \tag{A9}$$

$$|P_0^{\text{in}} - C_0| \leq \Delta_0,$$
$$C_0 = \sum_{k \in G_0} (L_k - P_k^{\text{PV}}), \Delta_0 = 0.08 \sum_{k \in G_0} B_k. \tag{A10}$$

The export limit constrains net power export. The energy-retention requirement sets the end-of-interval aggregate energy target to 101.5% of its initial value. The fault-recovery/load-priority task updates the feeder topology, inverter operating modes, and communication status while enforcing (A9). The combined task enforces (A9) and (A10), together with updated discharge permissions for selected resources.

TABLE A6
TOPOLOGY AND RESOURCE-CAPABILITY UPDATES

Item	Setting	Model update
33-bus topology	Open 6–7; close 8–21. Tie line: R = X = 2 Ω, 300 A.	Reconstruct the radial 33-bus admittance model
69-bus topology	Open 11–12; close 15–46. Tie line: R = 0.45 Ω, X = 0.35 Ω, 300 A.	Construct the radial 69-bus network with the specified tie line
Inverters and communication	Set unity power factor when k mod 5 = 0; communication is unavailable when k mod 11 = 1.	Update reactive capability and the admissible control domain
Combined-task permissions	Apply an additional discharge-permission restriction when k mod 7 = 1.	Update the storage active-power direction bounds

Graph-structure checks confirm connectivity and radiality for all 40 tasks. Branch exchange establishes the connected post-fault supply topology, and network constraints are reevaluated on the updated topology.

Resource indexing starts at $k = 0$. The tie branches and their ratings are specified operating assumptions for these experiments.

E.1 Task Thresholds and Natural-Language Specifications

Table A7 lists the parameters used to construct the four task types from each of the ten base states. Parameters in the same row apply to their respective tasks for that parent state. The priority-load shedding limit applies to both the fault-recovery/load-priority task and the combined task. Thresholds are reported in kW and kWh and are converted to MW and MWh in the execution model.

TABLE A7
STATE-SPECIFIC PARAMETERS OF THE NEW RULES

Buses	Parent state	Export limit (kW)	Energy minimum (kWh)	Shedding limit (kW)	Center C_0 (kW)	Half-width Δ_0 (kW)
33	001	0.14647	7.42735	0.10897	0.71479	0.41848
33	003	0.13842	6.71791	0.10237	0.66314	0.39548
33	005	0.14321	7.14876	0.11213	0.81051	0.40917
33	006	0.15736	5.25656	0.10955	0.61732	0.41898
33	009	0.13643	5.24603	0.09782	0.59218	0.35641
69	001	0.13676	5.23350	0.07423	0.11707	0.28026
69	003	0.14754	5.83825	0.07707	0.06606	0.27714
69	005	0.09894	4.03982	0.06214	0.25336	0.23086
69	006	0.11520	3.53902	0.06034	0.05479	0.22372
69	009	0.10802	4.16614	0.06798	0.27933	0.25204

Task specifications use three predefined forms: English with MW/MWh, English with kW/kWh, and Chinese with kW/kWh. These forms cover 16, 16, and 8 tasks, respectively. Each specification explicitly identifies the target, power-flow direction, units, and numerical requirements. Compiled rules are checked against the task definition for requirement coverage and parameter consistency.

Mapping a Centered-Band Requirement to Model Parameters

For parent state 003 of the 33-bus system, the combined task specifies a net-import band in terms of a center and a half-width. Group 0 net import must remain within ±0.395484100792 kW of 0.663141625668 kW, and total load shedding must not exceed 0.102366798922 kW. The compiled net-import interval is

$$0.267657524876 \leq P_0^{\text{in}} \leq 1.058625726460 kW. \tag{A11}$$

The center and half-width determine the lower and upper bounds used by the joint-correction skill. The English and Chinese specifications use the same power-flow sign convention. Unit conversion, verification of numerical literals, and rule-type validation precede execution.

The energy-retention task raises the end-of-interval aggregate energy target to 101.5% of its initial value and implements the new requirement through storage charging and discharging decisions. Export limits and centered bands constrain PCC exchanges during joint dispatch. The fault-recovery/load-priority task jointly updates topology, device capabilities, and the load-shedding limit. The four task types therefore assess the incorporation of market exchange limits, terminal energy requirements, post-fault capabilities, and combined operating requirements.

F. Benchmark Configurations and Evaluation Protocol

The core tasks compare the proposed method with AC nonlinear programming (AC NLP) and AC mixed-integer nonlinear programming (AC MINLP). The tasks with new requirements compare the proposed method with a fixed-weight graph neural network (GNN) followed by numerical repair. All methods use the same AC network, device constraints, objective parameters, and 90 s task budget. The GNN receives features representing current measurements, topology, capabilities, and control permissions. Numerical repair uses its established task model; the resulting actions are evaluated against the current operating requirements.

TABLE A8
METHOD CONFIGURATIONS AND ONLINE EXECUTION

Method	Execution configuration	Purpose
Proposed method: core tasks	Installed strategies in 54 tasks; zero control attains the zero-cost lower bound in six tasks; API disabled.	Assesses scaling of specialized execution under rules
Proposed method: new requirements	Online rule compilation; three strategy subagents propose plans in parallel; skills execute after contract validation.	Assesses translation of language requirements into task models and dispatch actions
AC NLP	Shared sparse elastic SQP backend; mode enumeration for up to six binary variables, bounded mode search otherwise.	Provides a continuous-optimization benchmark under common electrical constraints
AC MINLP	SCIP with a nonconvex branch-flow formulation and all PCC binary modes.	Provides a joint discrete–continuous optimization benchmark
Fixed GNN with repair	96 hidden units and eight graph message-passing layers; CPU inference followed by the established numerical repair.	Assesses response through existing feature and task interfaces

The GNN uses 588 labeled training instances and 120 labeled validation instances. Two architectures, with 64 hidden units and four layers and with 96 hidden units and eight layers, are each trained for 350 epochs using seed 4709. The 96-unit, eight-layer architecture is selected by validation loss. Total offline training time is 3073.38 s. Model weights remain fixed during online execution. The large language model (LLM) service identifier is deepseek-flash, with temperature = 0.2 and thinking disabled. Rule and strategy outputs are limited to 1200 and 600 tokens, respectively. At most two interface-validation rounds and two retries for transient transmission errors are allowed.

Task completion requires both independent feasibility verification of the original control action and satisfaction of the common cost-quality criterion. Let $J_{ref,i}$ denote the best-known feasible reference cost for task i. The cost deviation and the time of first attainment of the quality threshold are defined as

$$\delta_i(t) = \frac{J_i(t) - J_{ref,i}}{\max(|J_{ref,i}|, 0.1)} \leq 0.05,$$
$$t_i^Q = \inf\{t \in [0,90] : y_i(t) \in F_i^\varepsilon, \delta_i(t) \leq 0.05\}. \tag{A12}$$

The reference cost is the minimum cost among all independently verified feasible candidates. First attainment times are determined from candidate costs and their original timestamps. Tasks that have yet to meet the criterion at the budget limit are assigned a restricted time of 90 s. The mean is then calculated over the complete task set.

$$\tilde{t_i} = \min(t_i^Q, 90), T = \frac{1}{N}\sum_{i=1}^{N} \tilde{t_i}. \tag{A13}$$

Numerical computation uses a single thread. Online timing starts when the solver process is ready and includes feature computation, API calls, compilation, search, restoration, and online feasibility checks. Process initialization and model loading are recorded separately. The end of solution refinement and the first attainment of the quality threshold are timed separately.

G. Results as the Resource Population Increases

Table A9 reports results for all 60 core tasks by feeder and scale. Each feasibility count and quality-attainment count is out of ten tasks. The restricted means in (A13) include every task. Fig. A2 shows the effect of resource scaling under the same evaluation protocol.

TABLE A9
FEASIBILITY COUNTS, QUALITY-ATTAINMENT COUNTS, AND RESTRICTED MEAN TIMES

Buses	r	Proposed method Feasible / qualified; s	AC NLP Feasible / qualified; s	AC MINLP Feasible / qualified; s
33	1	10/10; 0.05	10/10; 0.06	10/10; 0.19
33	10	10/10; 2.38	10/10; 3.05	10/10; 2.75
33	100	10/10; 6.35	10/3; 64.73	10/9; 31.32
69	1	10/10; 0.08	10/10; 0.27	10/10; 0.41
69	10	10/10; 3.45	10/9; 27.60	10/10; 2.61
69	100	10/10; 21.73	10/2; 75.72	3/3; 80.29

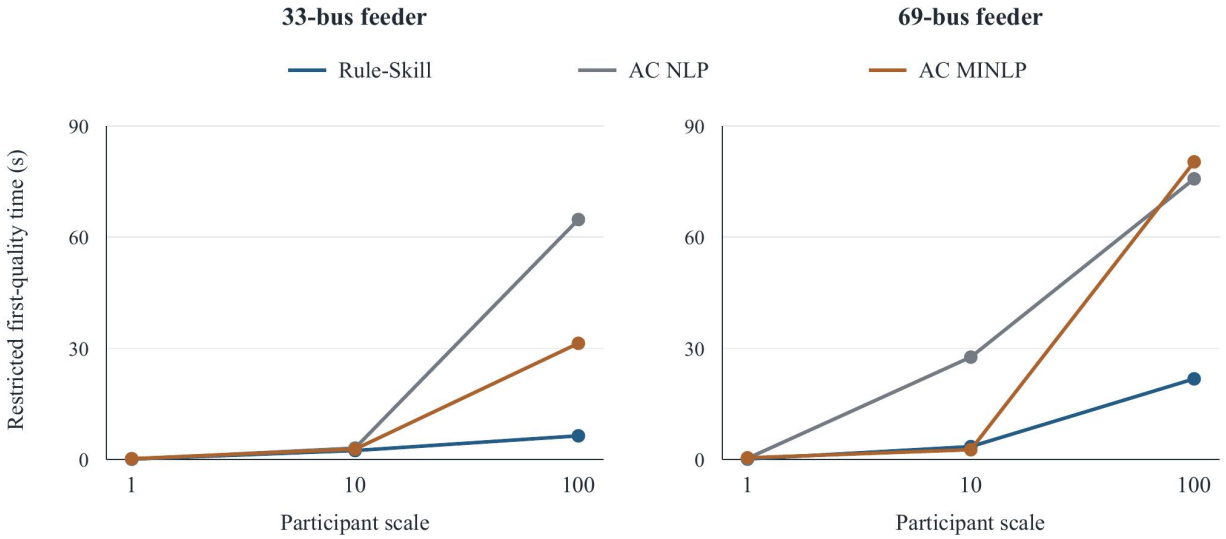


Fig. A2. Restricted mean time to first attainment of the quality threshold at three resource scales.

For the ten tasks on the 69-bus system with 900 PCC groups, the proposed method completes all tasks. Its mean first attainment time is 21.7308 s, with a maximum of 33.8200 s. AC NLP and AC MINLP attain the criterion in two and three tasks, respectively, with restricted mean times of 75.7166 and 80.2908 s. Across all 60 core tasks, the feasibility counts for the proposed method, AC NLP, and AC MINLP are 60, 60, and 53; their quality-attainment counts are 60, 44, and 52.

The scale-factor-100 cases comprise 20 tasks across the two feeders. Restricted mean times for the proposed method, AC NLP, and AC MINLP are 14.04, 70.22, and 55.81 s, respectively. A cluster bootstrap uses the physical parent state as the resampling unit, with 18 parent-state clusters, 5000 resamples, and seed 71039. The 95% confidence intervals for the mean time differences, defined as comparator time minus proposed-method time, are [43.96, 67.89] s for AC NLP and [29.73, 52.65] s for AC MINLP.

The scaling results characterize the ability of specialized execution paths to generate acceptable candidates under common electrical feasibility and cost-quality criteria. The installed strategies used for core tasks assess numerical execution efficiency, whereas online semantic compilation in the new-requirement tasks assesses task model adaptation. Together, these results support the rule–skill coordination mechanism described in the main text.

H. Task Completion under New Requirements and Physical Verification

TABLE A10
TASK COMPLETION UNDER NEW OPERATING REQUIREMENTS

Task type	Tasks	Proposed method Feasible / qualified	GNN with repair Feasible / qualified
Export limitation	10	10 / 10	0 / 0
Energy retention	10	10 / 10	0 / 0
Fault recovery / load priority	10	10 / 10	10 / 9
Combined requirements	10	10 / 10	0 / 0
Total	40	40 / 40	10 / 9

For all 40 tasks, the rules compiled by the proposed method are consistent with the task requirements. The mean first attainment time is 37.9143 s. Relative to the fixed GNN with its established repair procedure, the task-completion rate increases by 77.5 percentage points. Resampling the ten physical parent states yields a 95% confidence interval of [75.0, 82.5] percentage points. In the fault-recovery/load-priority group, both methods obtain feasible dispatches for all ten tasks, and their quality-attainment counts are ten and nine, respectively. These results indicate that the current topology and device-state interfaces support the electrical adjustments required in this group.

Electrical feasibility is verified independently from the original control actions. The nodal admittance matrix is reconstructed, AC power flow is solved by a Newton method in polar coordinates, and SOC, inverter capability circles, PCC exchanges, control permissions, and task constraints are evaluated individually. All 234 accepted candidates from the 100 tasks solved by the proposed method pass verification. The maximum nodal power-balance residual is 1.181×10^{-10} p.u. The same verification also confirms 123 final feasible candidates produced by the benchmark methods.

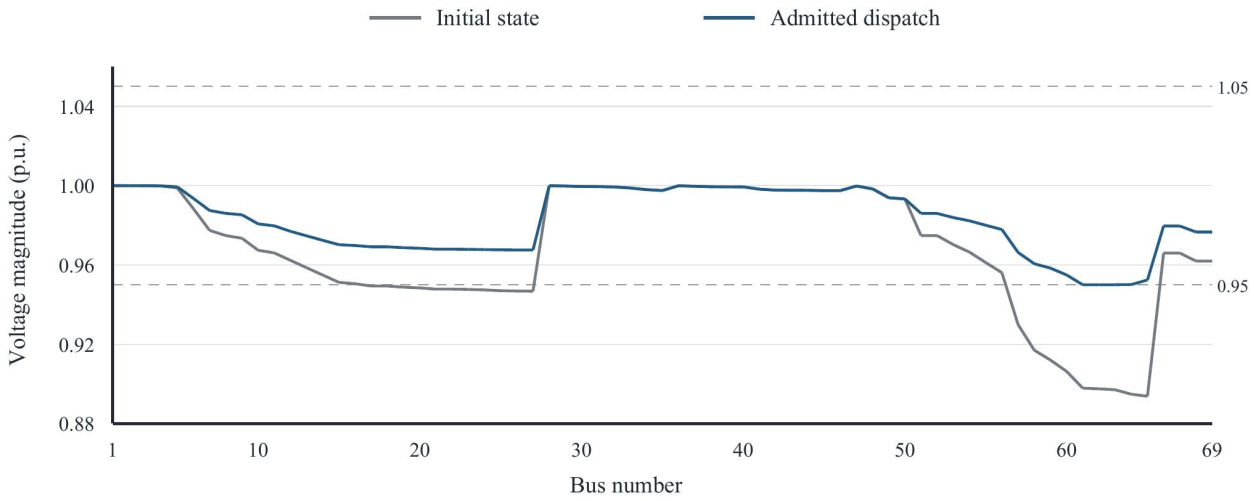


Fig. A3. Initial and post-dispatch bus voltages for the undervoltage scenario on the 69-bus system with 900 PCC groups.

Fig. A3 uses undervoltage scenario 001. The minimum initial bus voltage is 0.893795 p.u. and increases to 0.950000 p.u. after dispatch. The corresponding candidate is obtained at 33.6763 s. Bus voltages for all 234 accepted candidates satisfy the 0.95–1.05 p.u. limits within the specified numerical tolerances, and the maximum branch current-to-limit ratio is 0.994278. Electrical verification uses the actual topology and dispatch controls for each scenario.

Fig. A3 is recalculated from the accepted controls for this scenario. Cost quality is evaluated using the common criterion in (A12) and a 90 s time window.

I. Common Numerical Settings

Power-flow calculation, continuous optimization, discrete mode search, and final feasibility checks use the numerical settings in Table A11. All methods share the same 90 s online budget and physical-constraint tolerances.

TABLE A11
COMMON NUMERICAL SETTINGS

Item	Setting
Online budget / numerical threads	90 s / 1
Power-flow convergence and equality residual	Backward/forward-sweep convergence: 10^{-11}; independent Newton convergence: 10^{-10}; acceptance residual: 10^{-8} p.u.
Inequality and device-bound tolerances	Normalized network constraints: 10^{-7}; device powers, SOC, and independently checked task rules: 10^{-8} in their respective scales.
Shared continuous backend	Sparse elastic SQP: up to 50 iterations; Clarabel subproblems: up to 150 iterations, tolerance 10^{-9}.
MINLP settings	Single-threaded SCIP; feasibility tolerance: 10^{-9}; relative objective gap: 10^{-4}; online execution uses the remaining task budget.

References

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